

Week 1 Examine TCP/IP and OSI models in action

Report by Ezekiel Waweru CNS06-24049

Introduction

This was an assignment to help explore the more technical aspects of the TCP/IP and OSI models. Through the use of cisco packet tracer and an already configured web client and web server, one was able to use preselected protocols such as HTTP to see how packets are transmitted from start to finish in the OSI model.

Questions

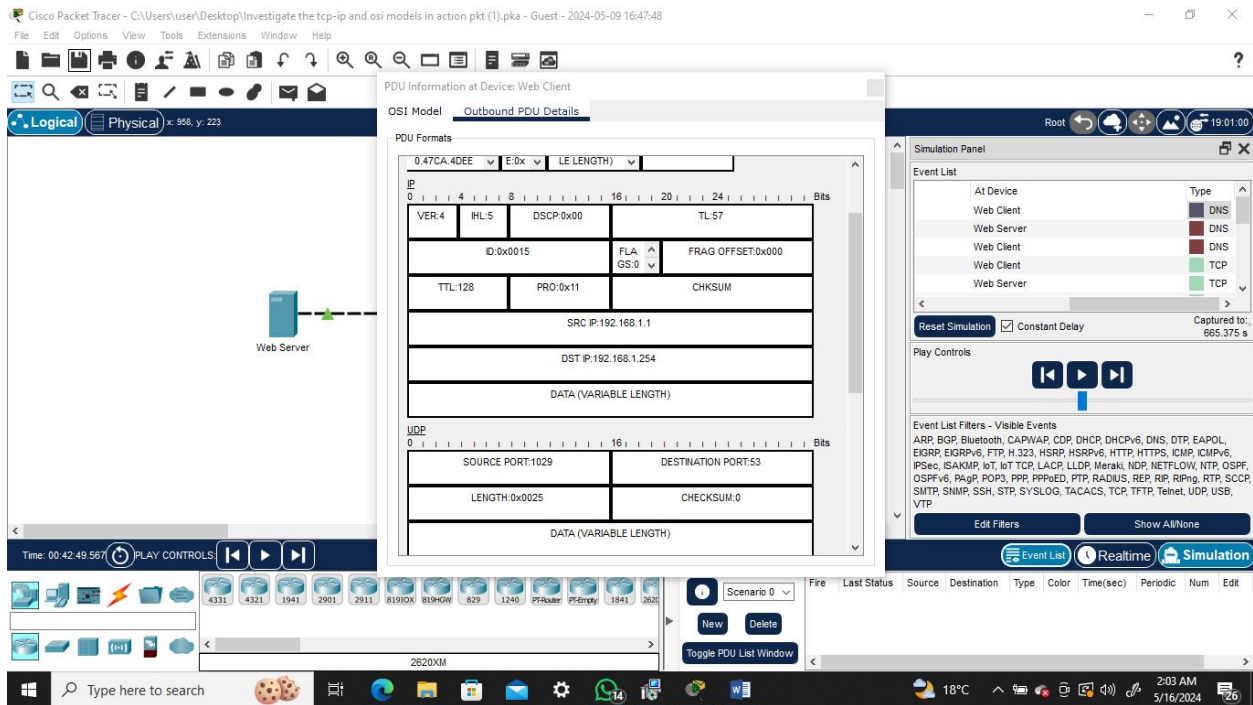
What information is listed in the numbered steps directly below the **In Layers** and **Out Layers** boxes for Layer 7? The DNS client sends a DNS query to the DNS server.

- I. What is the **Dst Port** value for **Layer 4** under the **Out Layers** column? 53
- II. What is the **Dest. IP** value for **Layer 3** under the **Out Layers** column? 192.168.1.254
- III. What information is displayed at Layer 2 under the **Out Layers** column?
 - The next hop IP address is a unicast. The ARP process looks it up in the ARP table
 - The next-hop IP address is in the ARP table. The ARP process sets the frame's destination MAC address to the one found on the table.
 - The device encapsulates the PDU into an Ethernet frame.

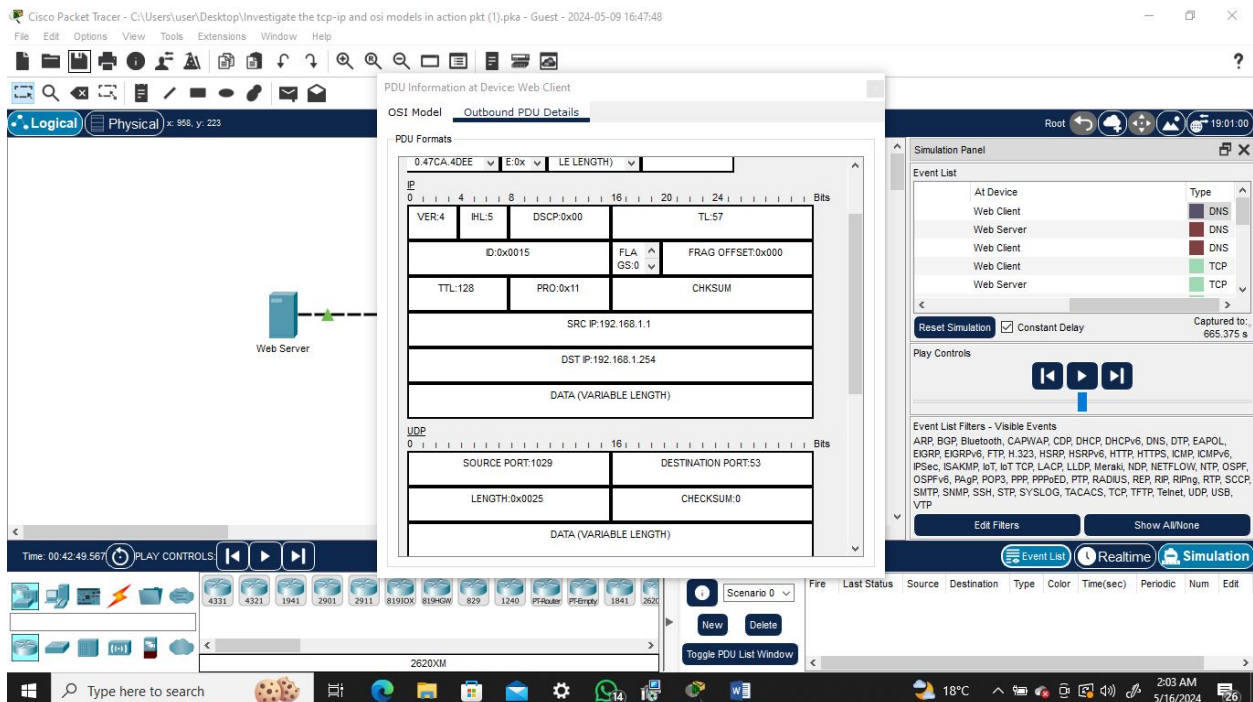
The screenshot displays the Cisco Packet Tracer interface. A 'PDU Information at Device: Web Client' window is open, showing the 'Outbound PDU Details' tab. The 'In Layers' column lists Layer 7 through Layer 1. The 'Out Layers' column lists Layer 7: DNS, Layer 6, Layer 5, Layer 4: UDP Src Port: 1029, Dst Port: 53, Layer 3: IP Header Src. IP: 192.168.1.1, Dest. IP: 192.168.1.254, Layer 2: Ethernet II Header 0060.47CA.4DEE >> 0001.96A9.401D, and Layer 1: Port(s): FastEthernet0. Below the layers, three numbered steps describe the ARP process: 1. The next-hop IP address is a unicast. The ARP process looks it up in the ARP table. 2. The next-hop IP address is in the ARP table. The ARP process sets the frame's destination MAC address to the one found in the table. 3. The device encapsulates the PDU into an Ethernet frame. The 'Simulation Panel' on the right shows the event list with DNS and TCP events. The 'Event List Filters' section lists various protocols like ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DNS, etc. The bottom status bar shows the time as 00:42:48.567 and the play controls.

- IV. What is the common information listed under the **IP** section of **PDU Details** as compared to the information listed under the **OSI Model** tab? With which layer is it associated? Layer 3

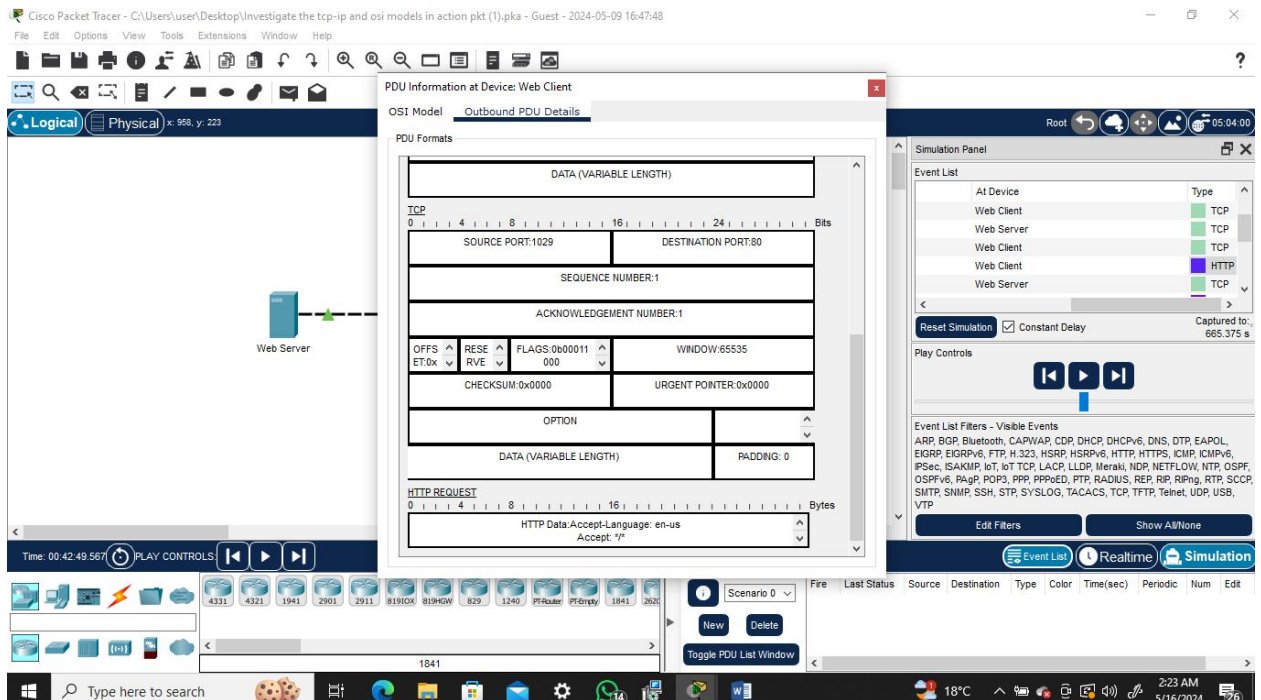
Checksum, Protocols such as Time to live



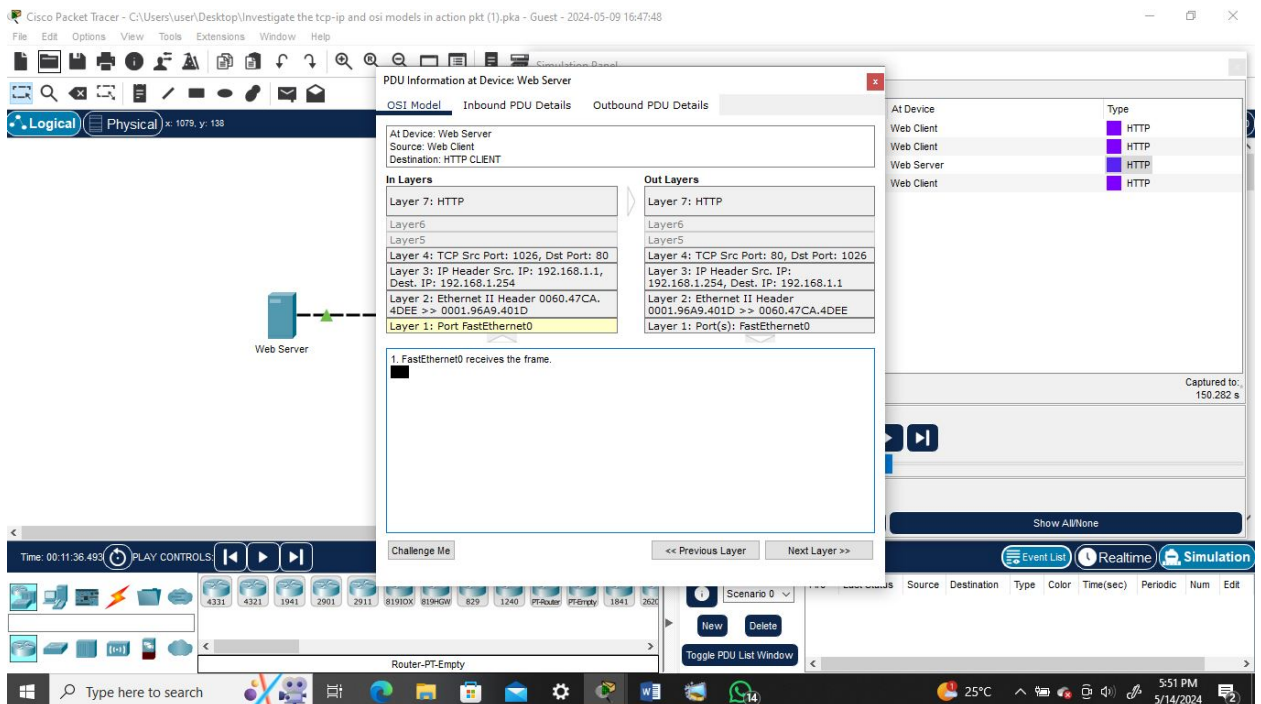
- V. What is the common information listed under the **TCP** section of **PDU Details**, as compared to the information listed under the **OSI Model** tab, and with which layer is it associated? Transport Layer Source and Acknowledgement number



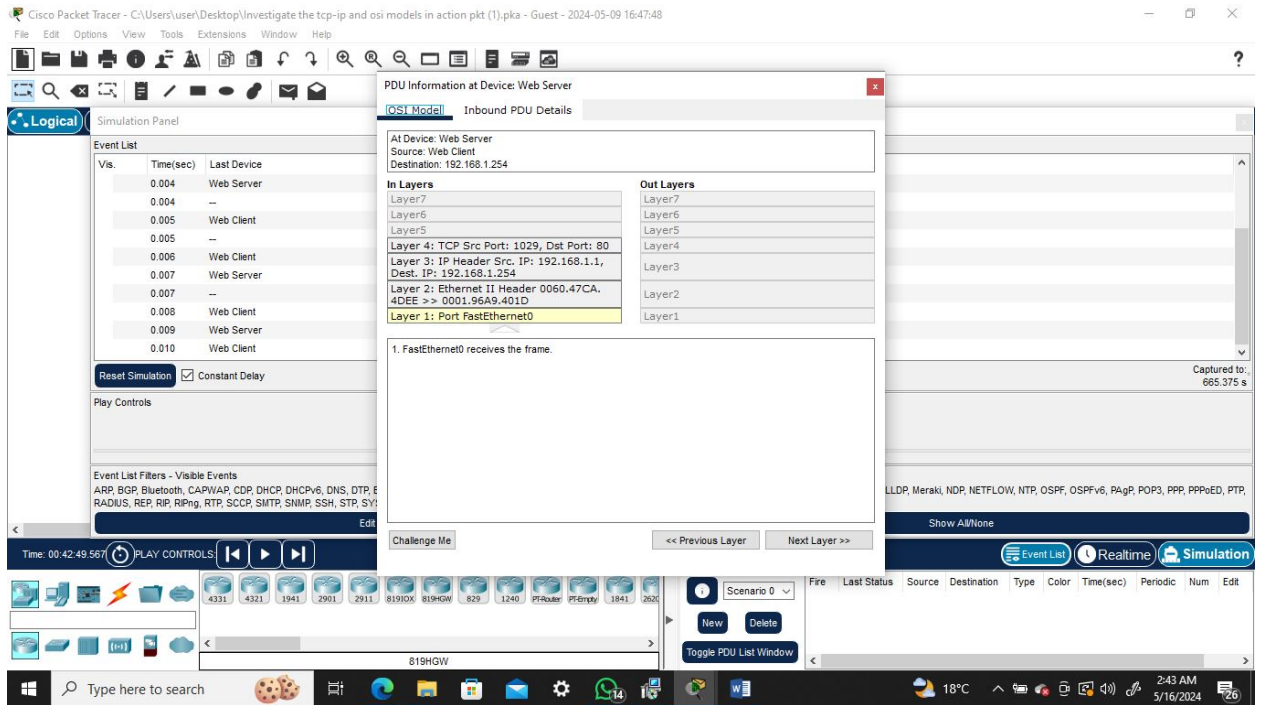
- VI. What is the **Host** listed under the **HTTP** section of the **PDU Details**? What layer would this information be associated with under the **OSI Model** tab? www.osi.local. It would be located in layer 7, Application layer



- VII. Comparing the information displayed in the In Layers column with that of the Out Layers column, what are the major differences? The In Layers column has all the data and none is in the Out layers column.



- VIII. How many tabs are displayed with this event? Explain. There are two tabs, The OSI Model tab and the PDU detail tab. This is because it is the onset of data transmission. If there were more events to be examined, then more tabs will be added



Part 2

In the **Event List Filters > Visible Events** section, click **Show All/None**.

What additional Event Types are displayed? 55 additional events

Click the first DNS event in the **Type** column. Explore the **OSI Model** and **PDU Detail** tabs and note the encapsulation process. As you look at the **OSI Model** tab with **Layer 7** highlighted, a description of what is occurring is listed directly below the **In Layers** and **Out Layers** ("1. The DNS client sends a DNS query to the DNS server."). This is very useful information to help

understand what is occurring during the communication process.

The screenshot shows the Cisco Packet Tracer interface. The Event List is visible, showing a sequence of events from 0.000 to 0.006. The last event at 0.006 is an HTTP request from a Web Client to a Web Server. The PDU Information window is open, showing the Outbound PDU Details for the selected event. The PDU is an HTTP request from the Web Client to the Web Server, with the destination IP address 192.168.1.254. The PDU is captured at the Web Client device.

Vis.	Time(sec)	Last Device	At Device	Type
Visible	0.000	--	Web Client	DNS
	0.001	Web Client	Web Server	DNS
	0.002	Web Server	Web Client	DNS
	0.002	--	Web Client	TCP
	0.003	Web Client	Web Server	TCP
	0.004	Web Server	Web Client	TCP
	0.004	--	Web Client	HTTP
	0.005	Web Client	Web Server	TCP
	0.005	--	Web Client	HTTP
	0.006	Web Client	Web Server	HTTP

PDU Information at Device: Web Client

OSI Model Outbound PDU Details

At Device: Web Client
Source: Web Client
Destination: 192.168.1.254

In Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer2
Layer1

Out Layers
Layer7: DNS
Layer6
Layer5
Layer4: UDP Src Port: 1029, Dst Port: 53
Layer3: IP Header Src. IP: 192.168.1.1, Dst. IP: 192.168.1.254
Layer2: Ethernet II Header 0060.47CA.4DEE >> 0001.96A9.401D
Layer1: Port(s): FastEthernet0

1. The DNS client sends an A DNS query to the DNS server.

Click the **Outbound PDU Details** tab.

What information is listed in the **NAME** field: in the DNS QUERY section? www.osi.local

The screenshot shows the Cisco Packet Tracer interface. The Event List is visible, showing a sequence of events from 0.000 to 0.006. The last event at 0.006 is an HTTP request from a Web Client to a Web Server. The PDU Information window is open, showing the Outbound PDU Details for the selected event. The PDU is an HTTP request from the Web Client to the Web Server, with the destination IP address 192.168.1.254. The PDU is captured at the Web Client device.

Vis.	Time(sec)	Last Device	At Device	Type
Visible	0.000	--	Web Client	DNS
	0.001	Web Client	Web Server	DNS
	0.002	Web Server	Web Client	DNS
	0.002	--	Web Client	TCP
	0.003	Web Client	Web Server	TCP
	0.004	Web Server	Web Client	TCP
	0.004	--	Web Client	HTTP
	0.005	Web Client	Web Server	TCP
	0.005	--	Web Client	HTTP
	0.006	Web Client	Web Server	HTTP

PDU Information at Device: Web Client

OSI Model Outbound PDU Details

PDU Formats

LENGTH: 0x0025 CHECKSUM: 0

DATA (VARIABLE LENGTH)

DNS Header

Transaction ID: 0xde4f QDCOUNT: 1 NSCOUNT: 0 QDCOUNT: 1 ANCOUNT: 0 NSCOUNT: 0 ARCOUNT: 0

DNS Query

NAME (VARIABLE LENGTH): www.osi.local

TYPE: 1 CLASS: 1

TTL: 86400

LENGTH: 0

Click the last DNS **Info** colored square box in the event list.

At which device was the PDU captured? Web Client

Cisco Packet Tracer - C:\Users\user\Desktop\Investigate the tcp-ip and osi models in action pkt (1).pka - Guest - 2024-05-09 16:47:48

File Edit Options View Tools Extensions Window Help

Logical Simulation Panel

Event List

Vis	Time(sec)	Last Device	At Device	Type
Visible	0.000	--	Web Client	DNS
	0.001	Web Client	Web Server	DNS
	0.002	Web Server	Web Client	DNS
	0.002	--	Web Client	TCP
	0.003	Web Client	Web Server	TCP
	0.004	Web Server	Web Client	TCP
	0.004	--	Web Client	HTTP
	0.005	Web Client	Web Server	TCP
	0.005	--	Web Client	HTTP
	0.006	Web Client	Web Server	HTTP

Reset Simulation ☒ Constant Delay

Play Controls

Event List Filters - Visible Events

ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IGRP, L2TP, L2TPv3, LLDP, LLDP-MED, MVRP, NTP, PIM, PIMv6, RIPv2, RIPv2v6, RIPv3, RIPv3v6, RSTP, RSTPv6, SLL, SLLv6, STP, STPv6, Syslog, Syslogv6, TACACS, TACACSv6, Telnet, Telnetv6, UDP, UDPv6, VTP

Time: 00:42:49.567 PLAY CONTROLS

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PDU Information at Device: Web Client

OSI Model Inbound PDU Details

At Device: Web Client
Source: Web Client
Destination: 192.168.1.254

In Layers

Layer	Protocol
Layer 7: DNS	
Layer 6	
Layer 5	
Layer 4: UDP Src Port: 53, Dst Port: 1029	
Layer 3: IP Header Src: IP: 192.168.1.254, Dest: IP: 192.168.1.1	
Layer 2: Ethernet II Header 0001.96A9.401D >> 0060.47CA.4DEE	
Layer 1: Port FastEthernet0	

Out Layers

Layer	Protocol
Layer 7	
Layer 6	
Layer 5	
Layer 4	
Layer 3	
Layer 2	
Layer 1	

1. FastEthernet0 receives the frame.

Challenge Me

<< Previous Layer Next Layer >>

New Delete

Toggle PDU List Window

What is the value listed next to **ADDRESS:** in the DNS ANSWER section of the **Inbound PDU Details**? 192.168.1.254

Cisco Packet Tracer - C:\Users\user\Desktop\Investigate the tcp-ip and osi models in action pkt (1).pka - Guest - 2024-05-09 16:47:48

File Edit Options View Tools Extensions Window Help

Logical Simulation Panel

Event List

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Visible	0.000	--	Web Client	DNS
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	0.002	Web Server	Web Client	DNS
	0.002	--	Web Client	TCP
	0.003	Web Client	Web Server	TCP
	0.004	Web Server	Web Client	TCP
	0.004	--	Web Client	HTTP
	0.005	Web Client	Web Server	TCP
	0.005	--	Web Client	HTTP
	0.006	Web Client	Web Server	HTTP

Reset Simulation ☒ Constant Delay

Play Controls

Event List Filters - Visible Events

ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IGRP, L2TP, L2TPv3, LLDP, LLDP-MED, MVRP, NTP, PIM, PIMv6, RIPv2, RIPv2v6, RIPv3, RIPv3v6, RSTP, RSTPv6, SLL, SLLv6, STP, STPv6, Syslog, Syslogv6, TACACS, TACACSv6, Telnet, Telnetv6, UDP, UDPv6, VTP

Time: 00:42:49.567 PLAY CONTROLS

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PDU Information at Device: Web Client

OSI Model Inbound PDU Details

PDU Formats

Field	Value
NSCOUNT:0	ARCOUNT:0
DNS Query	
NAME (VARIABLE LENGTH): www.osi.local	
TYPE:1	CLASS:1
TTL:66400	
LENGTH:0	
DNS Answer	
NAME (VARIABLE LENGTH): www.osi.local	
TYPE:1	CLASS:1
TTL:66400	
LENGTH:4	IP:192.168.1.254

Find the first **HTTP** event in the list and click the colored square box of the **TCP** event immediately following this event. Highlight **Layer 4** in the **OSI Model** tab.

In the numbered list directly below the **In Layers** and **Out Layers**, what is the information displayed under items 4 and 5? The TCP connection is successful, The device sets the connection state to ESTABLISHED

Simulation Panel

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Visible	0.000	--	Web Client	DNS
	0.001	Web Client	Web Server	DNS
	0.002	Web Server	Web Client	DNS
	0.002	--	Web Client	TCP
	0.003	Web Client	Web Server	TCP
	0.004	Web Server	Web Client	TCP
	0.004	--	Web Client	HTTP
	0.005	Web Client	Web Server	TCP
	0.005	--	Web Client	HTTP
	0.006	Web Client	Web Server	HTTP

Reset Simulation ☒ Constant Delay

Play Controls

Event List Filters - Visible Events

ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSEC, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Time: 00:42:49.567

Router-PT-Empty

PDU Information at Device: Web Server

OSI Model Inbound PDU Details

At Device: Web Server
Source: Web Client
Destination: 192.168.1.254

In Layers

- Layer7
- Layer6
- Layer5
- Layer4: TCP Src Port: 1029, Dst Port: 80
- Layer3: IP Header Src. IP: 192.168.1.1, Dst. IP: 192.168.1.254
- Layer2: Ethernet II Header 0060.47CA.4DEE >> 0001.96A9.401D
- Layer1: Port FastEthernet0

Out Layers

- Layer7
- Layer6
- Layer5
- Layer4
- Layer3
- Layer2
- Layer1

1. The device receives a TCP ACK segment on the connection to 192.168.1.1 on port 1029.
2. Received segment information: the sequence number 1, the ACK number 1, and the data length 20.
3. The TCP segment has the expected peer sequence number.
4. The TCP connection is successful.
5. The device sets the connection state to ESTABLISHED.

Click the last TCP event. Highlight Layer 4 in the **OSI Model** tab. Examine the steps listed directly below **In Layers** and **Out Layers**.

What is the purpose of this event, based on the information provided in the last item in the list (should be item 4)?

Ezekiel Waweru CNS06-24049 - Word

File Home Insert Design Layout References Mailings Review View Tell me what you want to do...

Clipboard Font Paragraph Styles Editing

Simulation Panel

Vis.	Time(sec)	Last Device	At Device	Type
Visible	0.000	--	Web Client	DNS
	0.001	Web Client	Web Server	DNS
	0.002	Web Server	Web Client	DNS
	0.002	--	Web Client	TCP
	0.003	Web Client	Web Server	TCP
	0.004	Web Server	Web Client	TCP
	0.004	--	Web Client	HTTP
	0.005	Web Client	Web Server	TCP
	0.005	--	Web Client	HTTP
	0.006	Web Client	Web Server	HTTP

Reset Simulation ☒ Constant Delay

Play Controls

Event List Filters - Visible Events

ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSEC, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Time: 00:42:49.567

Router-PT-Empty

PDU Information at Device: Web Server

OSI Model Inbound PDU Details

At Device: Web Server
Source: Web Client
Destination: 192.168.1.254

In Layers

- Layer7
- Layer6
- Layer5
- Layer4: TCP Src Port: 1029, Dst Port: 80
- Layer3: IP Header Src. IP: 192.168.1.1, Dst. IP: 192.168.1.254
- Layer2: Ethernet II Header 0060.47CA.4DEE >> 0001.96A9.401D
- Layer1: Port FastEthernet0

Out Layers

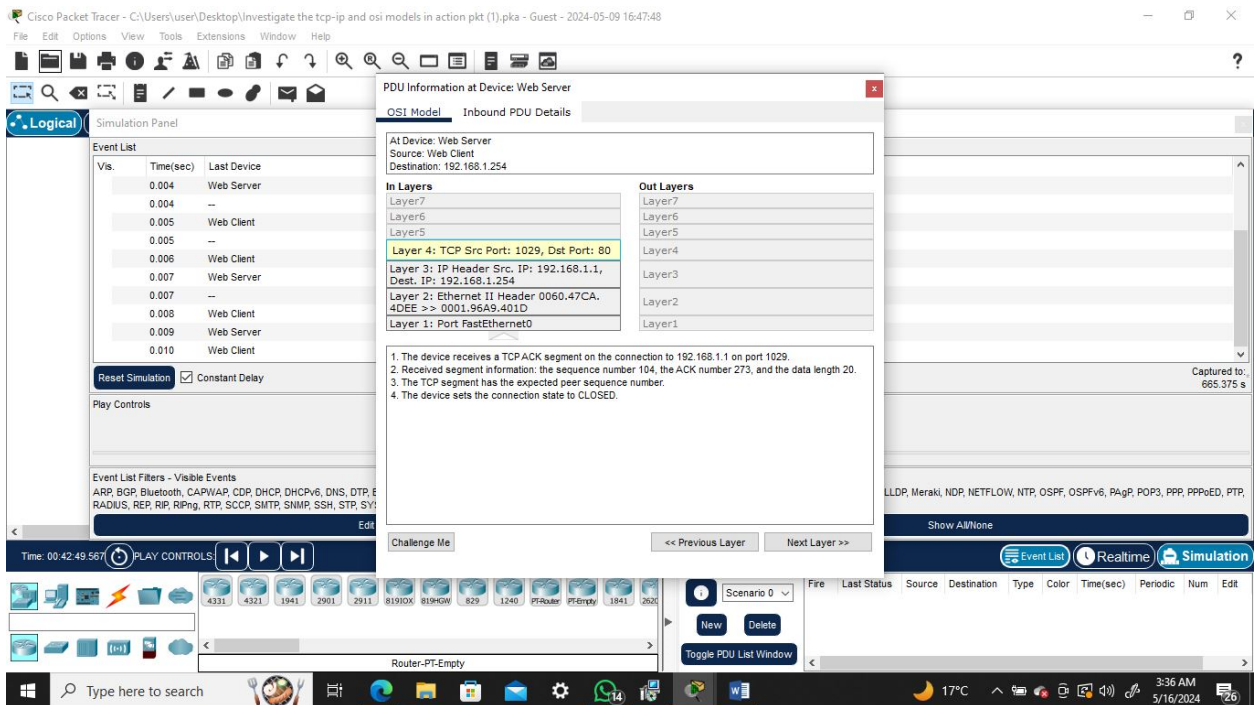
- Layer7
- Layer6
- Layer5
- Layer4
- Layer3
- Layer2
- Layer1

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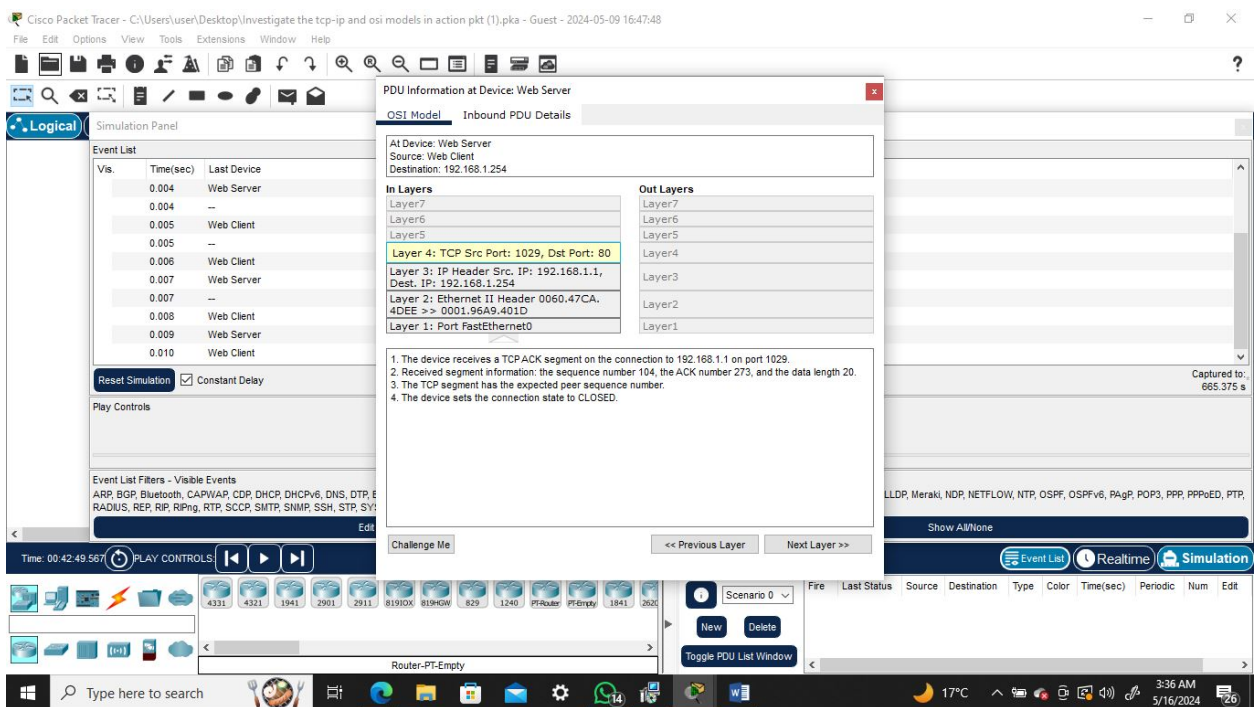
Click the last TCP event. Highlight Layer 4 in the **OSI Model** tab. Examine the steps listed directly below **In Layers** and **Out Layers**.

What is the purpose of this event, based on the information provided in the last item in the list (should be item 4)?

Based on the information that was inspected during the Packet Tracer capture, what port number is the **Web Server** listening on for the web request? Port 1029



What port is the **Web Server** listening on for a DNS request? 80



Conclusion

Through the use of Cisco Packet tracer, this assignment became possible. The goal of the assignment was to give an overview of what happens to data in transit from the first OSI layer to the last. Protocols such as HTTP were used as they help in converting domain names into

IP addresses. This helps one get the information they were seeking when they carry out a web search. Other ports such port 80 which used for HTTP and others like 1029 were widely used to showcase communication from source to destination. It was a good assignment which has helped build skill.